

As research and development (R&D) in this area progresses, this technology will soon affect everything we do in our daily lives, from communicating to conducting our finances. Quantum computing, with all its advantages, is attractive to different areas of technology including autonomous vehicles, weather forecasting, transit planning, pharmaceutical research, secure financial communications, resource exploitation, and sensing. For an air carrier, it could assist with delays, rerouting, and rescheduling of flights that have been cancelled, adding to all this greater security while paying bills online. This advancement even allows more specific medical imaging systems to be created.

The materials used in nanoscale quantum computers have unique characteristics that could increase the sophistication of model chemical processes and material design. This would help designers and engineers develop better drugs, batteries, and advanced materials with superior electrical and mechanical properties.

Even with such huge advantages possible, the quantum field is not yet ready for commercialisation due to significant scientific and technological difficulties involved. Innovative and high-impact applications might be developed by companies based on substantial and dependable government backing, which could lead to economic development and employment. These economic outcomes would act as feedback for a developing quantum economy.

Fundamental research, translation, technological development, and solving problems related to national objectives must be included in our initiatives. This inclusion may involve quantum computers, quantum communication, indigenous capacity, and trained personnel needed for quantum based research.

Industrial revolution 4.0

We only have a basic understanding of development, but a well-rounded approach must be used, with all of the global stakeholders' participation involving not only the public and commercial sectors but also academia and civil society. Let us look at how the industries have evolved

through the revolution.

Water and steam power helped propel the first industrial revolution, which included automation. The second revolutionised mass manufacturing with the use of electricity. The third revolution used electronics and information technology in the manufacturing process. The fourth industrial revolution is in the process of being built with the third industrial revolution as its foundation. It began in late 20th century with the digital revolution. Convergence is helping the emerging technologies move seamlessly across the physical, digital, and biological worlds.

There are three key factors that differentiate the current transition era from previous ones: pace, scope, and system effect. Historical precedent can no longer be applied to the present pace of breakthroughs. While earlier industrial revolutions took place at a linear speed, the fourth industrial revolution is progressing at an exponential rate. Further, it is interrupting almost every sector and nation across the world, causing drastic changes from our methods of manufacturing, our management, to our whole system of governance.

We believe that mobile devices—with all their processing power, storage capacity, and access to information—will open the doors to a variety of applications for billions of people. In addition, advancements in areas such as artificial intelligence, robots, the Internet of Things, autonomous cars, 3D printing, nanotechnology, biotechnology, materials science, energy storage, and quantum computing will have a positive impact on the potential solutions.

Frontier research

Quantum technology offers an enormous potential such as secure, unhackable communication lines, using which single photon emitters and detectors may be developed, which may aid in averting cyberattacks that damage a nation's infrastructure.

Google, Microsoft, and IBM are using quantum technology to boost their computing businesses. A major reason for the global energy problem is the enormous amount of energy needed to maintain the increasing

Internet usage. Modernising our computing components and communications equipment helps in protecting the environment. So a new generation of quantum devices is being developed at Technion to improve computers and communications.

With the help of quantum technology, we can create sensing devices that use smaller, more sensitive sensors. The diagnostic sensors and super-resolution medical imaging applications of these materials may be utilised for several reasons including implanted sensors and medical imaging. Ultra-sensitive night vision technology and sensors for magnetic fields, acceleration, temperature, hazardous chemicals, and harmful environmental conditions are being developed in tandem with the new technology.

Though all materials are quantum, experimentally we can harness the effects of quantum phenomena such as superconductivity, topological insulators, etc. Many of the quantum information system (QIS) materials use ion traps, defects, quantum dots, superconductors, and topological materials. Quantum materials, since they have a great deal of promise for QIS applications, would be an excellent addition to the Materials Genome Initiative.

Communication and cryptography

In an area of applied quantum physics strongly linked to quantum information processing and quantum teleportation, quantum communication may be seen as a subfield of quantum physics. The most intriguing use of quantum cryptography is in securing information routes against eavesdropping. Quantum key distribution (QKD) is the most well-known and is a developed application of quantum cryptography. It discusses the use of quantum phenomena to accomplish cryptographic tasks or to breach cryptographic systems.

The working concept of a QKD system is simple. Consider a scenario where two people, namely, Alice and Bob, utilise randomly polarised single photons to send a succession of random number sequences, all of which are used as cryptographic keys. The quantum channel connects the two stations, whereas the classical channel links them together. Alice uses